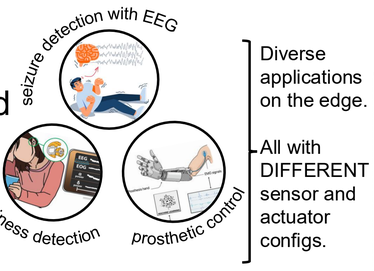
HiVec: Scalable and Energy Efficient CGRA in a RISC-V SoC for Wearables

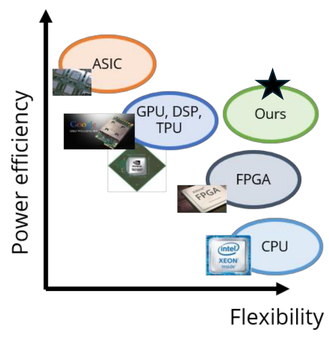
Rakshith Harish[2], Rohan Juneja[1], Pranav Dangi[1], Zhenyu Bai[1], Vishnu P. Nambiar[2], Yi Sheng Chong[2], Bin Zhao[2], Vishruti Ranjan[1], Rahul Dutta[2], Li-Shiuan Peh[1], Tulika Mitra[1], Anh Tuan Do[2]

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Multisensory Edge Analytics with HiVec

- The proliferation of "smart" wearable devices with sophisticated sensing/inference algorithms has created an exigency for fast, energy efficient compute.
- No single chip can handle all kinds of applications, and chip design cycles/fabrication is slow, high NRE.
- Coarse-grained reconfigurable arrays (CGRAs) deliver high energy efficiency with high programmability.
- The HiVec CGRA attains **16.2TOPS/W @ 0.4V and 267 MHz**, advancing SOTA with 7.3x higher normalized energy efficiency.





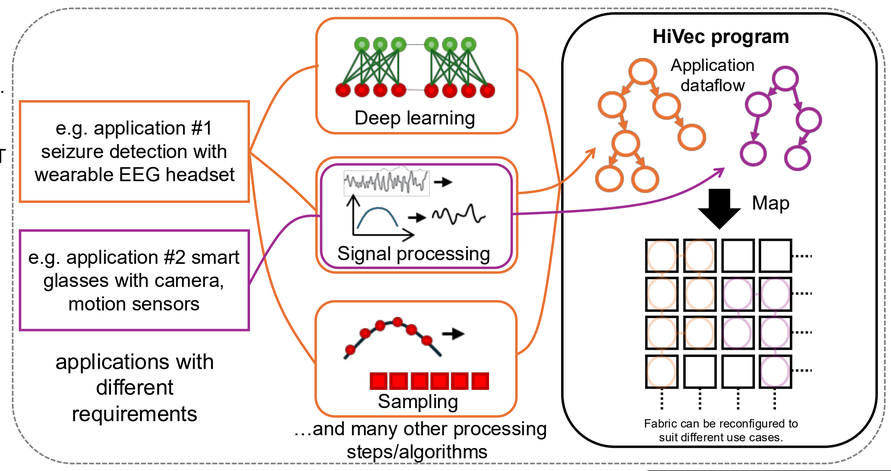
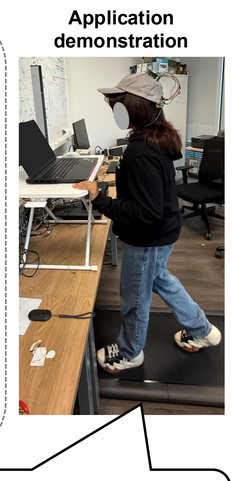


Fig 1: Kernels are converted into data flow graphs and mapped onto the hardware with high performance and energy efficiency.



Discreet wearable EEG with HiVec chip
Non-bulky, small footprint
DNN for **low latency BCI** tasks, Vision Transformer
Performance: **16.2 TOPS/Watt**

Decoupled Access Execute

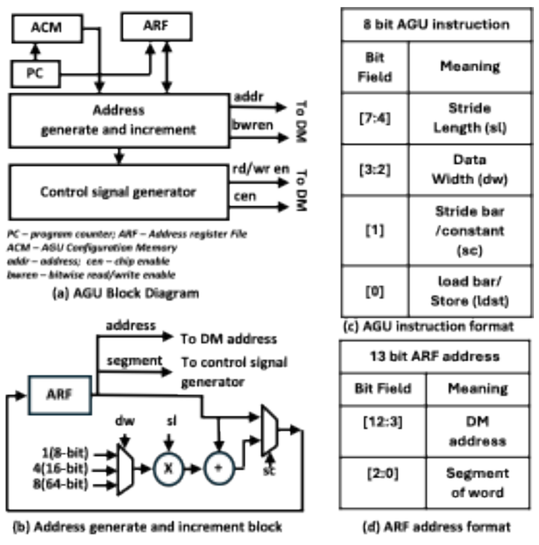


Fig 2: Microarchitecture of Decoupled Access Execute (compiler managed on chip memory)

Configurable Scalar/Vectorization support

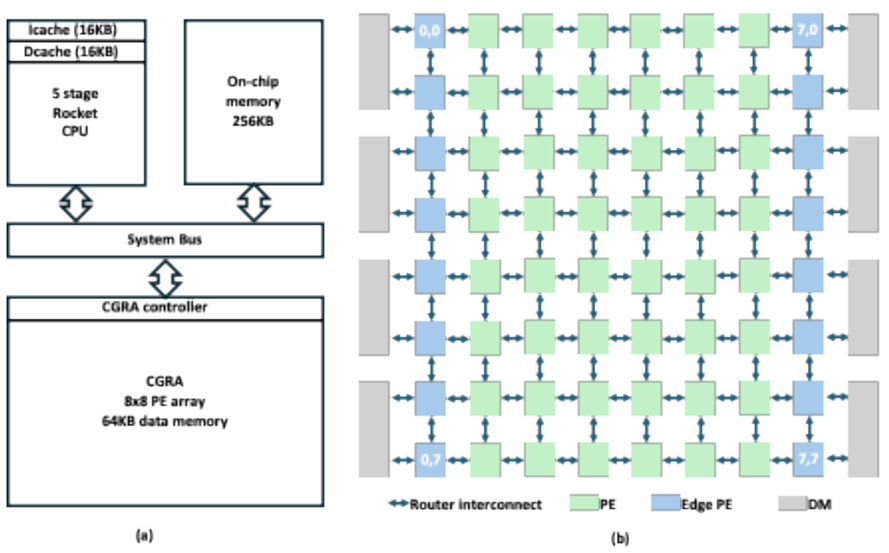


Fig 3: Microarchitecture of PE array

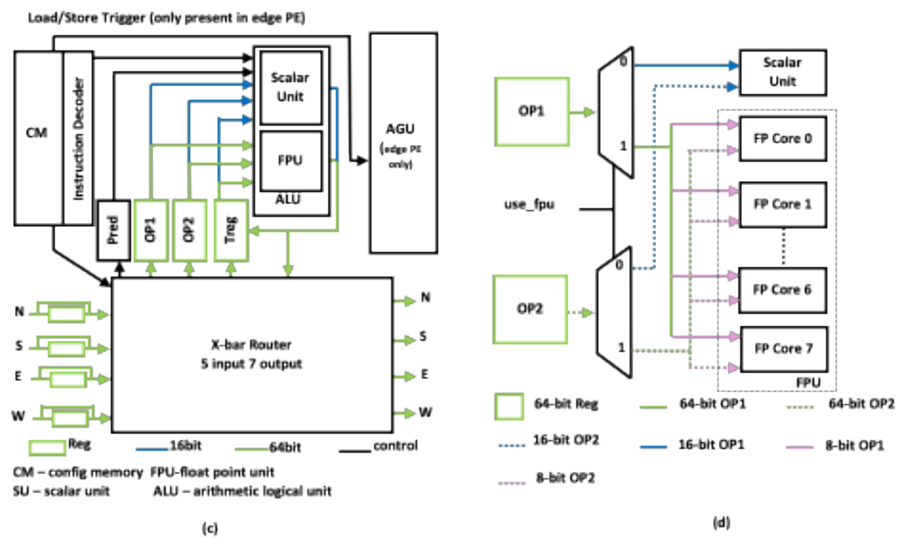


Fig 4: Microarchitecture of PE with configurable scalar, SIMD4, SIMD8 support at fp8, int16 precisions

Hierarchical, Modular Compilation & PnR

- Hierarchical compilation of Cpp, Triton, Helion programs/kernels onto the spatial dataflow fabric
- Multi-level Arch Representation serves as an ADL for portability, design space exploration
- Place-and-route mapping intra-core provides a performance model to the inter-core dataflow modeling
- Automatic generation of test data/layouts and simulator-in-the-loop for quick validation of mapping

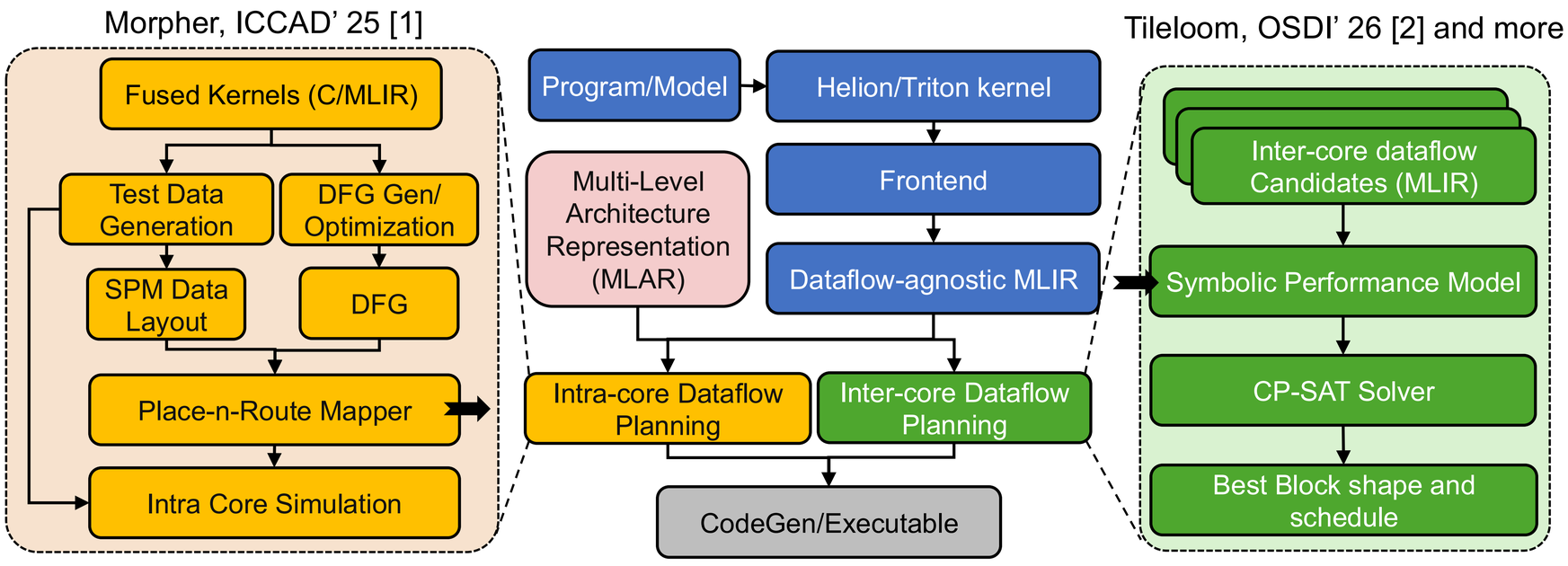


Fig. 5 End-to-end compiler toolchain

[1] R. Juneja et al., "Building an Open CGRA Ecosystem for Agile Innovation," 2025 IEEE/ACM ICCAD, Munich, Germany, 2025
[2] Wei Li et al., "TileLoom: Automatic Dataflow Planning for Tile-Based Languages on Spatial Dataflow Accelerators", 2026 Usenix OSDI, Seattle, Washington, 2026

Performance evaluation

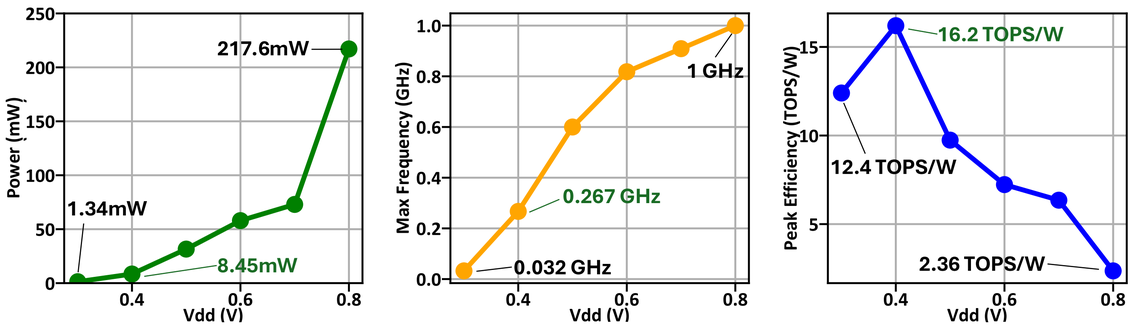


Fig. 6 CGRA metrics across V_{DD}, (a) power, (b) maximum frequency, and (c) peak energy efficiency.

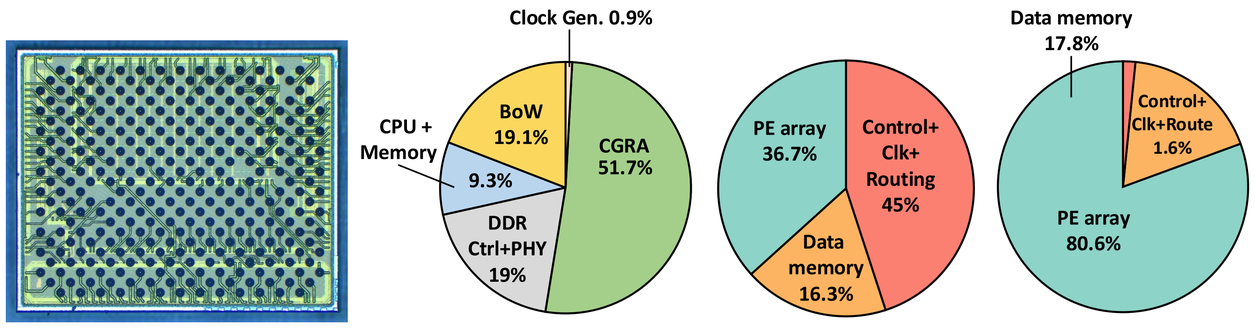


Fig. 7 Physical implementation: (a) chip micrograph, (b) SoC area breakdown, (c) area and (d) power breakdown of the CGRA

	ISSCC'21	DATE'20	DSD'24	SOCC'25	JSSCC'24	ISSCC'26	HiVec
Tech Node	16	28	45	45	16	3	**12**
Frequency (Ghz)	1.44	0.05	0.1	0.7	955	0.456	**1@0.8V 0.267@0.4V**
Peak Performance (GOPS)	388.4	NA	2.845	1.25	367@1.29V	2520	**512@0.8V 0.136@0.4V**
Peak Efficiency (GOPS/W)	209.5@0.55V	224	519	156	538@0.84V	28500	**2416@0.8V 16200@0.4V**
Normalised Efficiency (GOPS/W)	372.4	1219.5	1744	2193	956.4	1087	**16200**
Area (mm2)	4.4	0.174	0.2	0.013	20.1	0.338	**0.391**
Normalised Area (mm2)	2.93	0.174	0.0256	0.0016	2.048	1.86	**0.391**
Normalised Compute Density (TOPS/mm2)	0.125	NA	0.1109	0.781	0.179	1.36	**0.04 - 1.309**

Fig. 8 Performance comparison with state-of-the-art

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